



KILLSWITCH

Reversing the economic death spiral of modern air defense.

in collaboration with Group Sentinels

KILLSWITCH

SCENARIO



AIR DEFENSE TODAY

C-UAS Drone



- Range: short, (100-200m)
- Speed: slow, <5 km/h
- Quantity: high
- Destruction: 1-1 or 1-2

Laser Network

Low Power

Missile



- Range: <3km
- Speed: light speed
- Quantity: unlimited
- Destruction: 1-X

- Range: long, >50 km
- Speed: slow, >100 km/h
- Quantity: low
- Destruction: 1-1

THE PROBLEM

Modern militaries are bankrupting themselves to defend against cheap technology.



1. The Cost Disparity: Defense forces are firing \$3,000,000 kinetic interceptors to shoot down \$2,000 commercial fiber-optic drones.
2. The Human Bottleneck: Traditional systems require a 1:1 operator-to-weapon ratio. When facing a 40-drone swarm, a 1:1 human decision cycle is mathematically too slow to prevent perimeter breaches.
3. The Magazine Depth Flaw: Swarms do not present a single point of failure. Current air defense systems rely on a finite number of missile tubes; once the magazine is empty, the airspace is completely defenseless.

THE SOLUTION



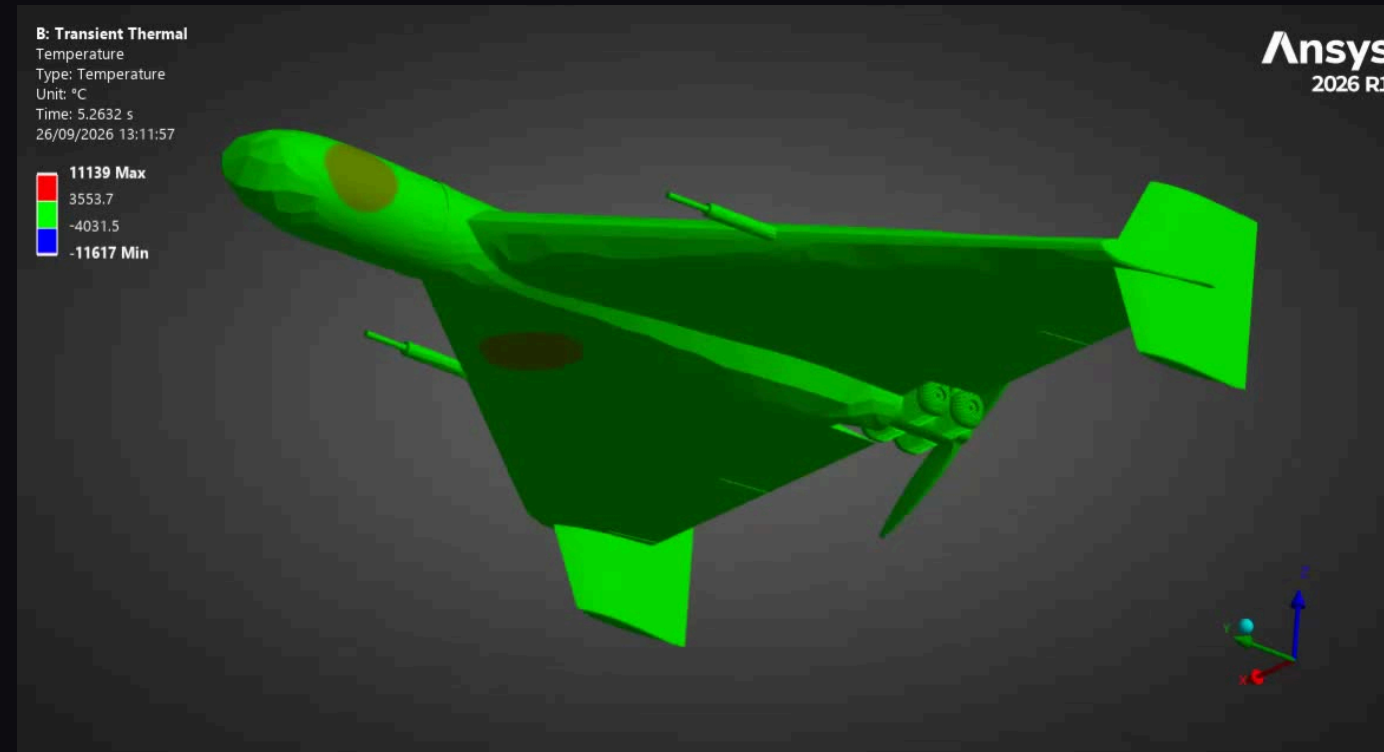
1. Imbalance Over Vaporization: Legacy military lasers (50kW+) require massive diesel generators just to melt metal.

KILLSWITCH utilizes low-cost, low-power (5kW) commercial lasers to apply steady thermal load to aerodynamic weak points (rotors, polymer struts). At high RPMs, a mere 5% structural deformation causes catastrophic aerodynamic imbalance, tearing the drone apart in mid-air.

2. 1:10 Decision Layer: Partnering with Team Sentinels, our software architecture eliminates the 1:1 manpower bottleneck, allowing a handful of operators to run the entire grid.

3. Human-in-the-Loop (HITL): The network automatically detects, tracks, and assigns the optimal overlapping lasers to the target. The machine handles the micro-millimeter tracking; the human operator retains ultimate firing authority, strictly satisfying military Rules of Engagement (ROE).

SIMULATION



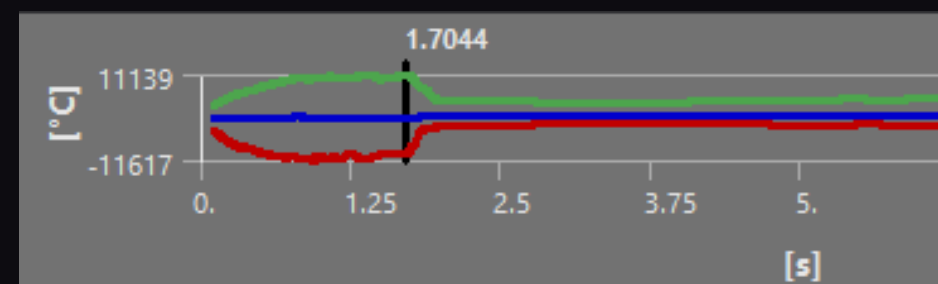
Threat: 100 simulated Shahed-class microdrones inbound at 50 m/s (180 km/h).

Convergence Math: 1 Laser = Baseline TTK. 2 Lasers = 50% TTK. 4 Lasers = Instant structural failure.

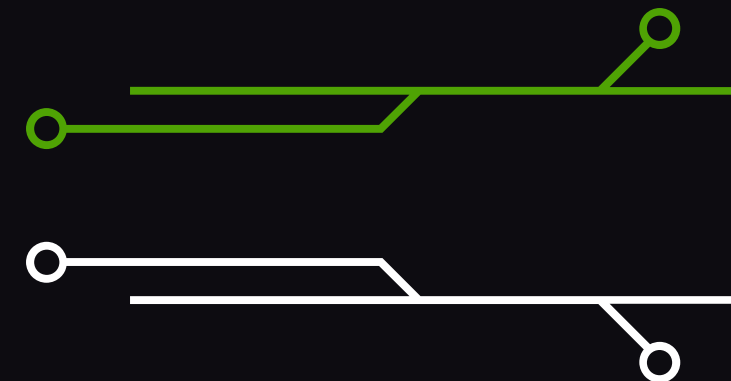
Result: Multi-node simultaneous engagement clears the swarm before perimeter breach.

COST EFFECTIVENESS:

<\$1 / node : <\$3 to take down a drone



- Works alongside kinetic interceptors
- 5 KW turrets thermally degrades a singular small UAV in 1.7s



HOW IT WORKS



A self-scaling, distributed defensive web that multiplies lethality through overlapping sectors.

Add a subheading



Detect: LiDAR acquires the target at a 3 km range.



Classify: Edge AI identifies the drone type and maps its specific structural weak points (e



Converge: Multiple networked lasers are assigned to overlap on the exact



Deconflict: The AI algorithm calculates safe firing trajectories to ensure zero collateral



Imbalance Kill: Thermal load warps the composite propeller. At high RPMs, the aerodynamic



OUR TEAM



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